

Executive Research Brief: Inverse-FNO-Damage

Physics-Constrained Structural Damage Inversion, Observability Limits, and Neural Learnability

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Field: Scientific Machine Learning & Structural Dynamics | **Status:** Complete Research Program (74 Unit Tests Passing) |
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1. Research Overview

This project investigates an ill-posed inverse problem in structural mechanics: identifying localized member damage (stiffness reduction δ in $[0, 0.5]^E$) in a building frame based purely on sparse, noisy acceleration and strain time-series collected during earthquake ground motion.

The core challenge is that civil structures possess nominal lateral symmetry. Localized stiffness degradation in laterally symmetric structural members (e.g. ground-floor left column vs. right column) generates horizontal floor vibrations that differ by **less than 0.14%**. Linearized sensitivity SVD shows that the bilateral difference unit vector $\mathbf{v}_{AB}$ aligns almost perfectly with the smallest singular vector of horizontal floor sensing ($|\angle \mathbf{v}_E, \mathbf{v}_{AB}| = 0.9982^\circ$). Under conventional horizontal floor accelerometers, bilateral failure modes lie inside an empirical near-null space.

2. Three Core Discoveries

Finding 1: Physical Observability Under Horizontal Sensing is Near-Null

Under calibrated 2% sensor noise, noise-whitened directional Fisher sensitivity for horizontal floor sensing is negligible ($\sqrt{I_{AB}} = 5.62$). In held-out bilateral test evaluations ($N = 30$), models trained on horizontal floor accelerometers achieve exactly **50.0% attribution accuracy (pure chance, $p = 1.000$)**. Multimodal sensing (\$4: horizontal + vertical + column axial strain) amplifies Fisher sensitivity by $404\times$ to $\sqrt{I_{AB}} = 2270.54$, elevating attribution accuracy to **90.0% (\$27/30, $p = 9.0 \times 10^{-6}$, 95% CI [73.5%, 97.9%])**.

Finding 2: Physical Observability is Necessary but Insufficient for Learnability

Column axial strain gauges (\$2) amplify directional Fisher sensitivity by **239 $\times$** over horizontal floor sensing ($\sqrt{I_{AB}} = 1344.18$), because axial deformation directly captures lateral rocking load transfer. Standard theory suggests the inverse network should easily distinguish the states.

However, in practice, models trained on \$2 achieve only **50.0% attribution accuracy (\$15/30, $p = 1.000$)**. Tracking gradient norms reveals that horizontal floor accelerations ($\sim 1 \text{ m/s}^2$) account for **over 98%** of the total backpropagation gradient norm, drowning out micro-strain gradients ($\sim 10^{-5} \text{ m/m}$). The optimizer settles into a symmetric local minimum near the mean before strain signals can guide the weights. *High Fisher information does not guarantee neural learnability under multiscale gradient dynamics.*

Finding 3: Cross-Structure Direction–Magnitude Decoupling

When evaluated across 530 OpenSees dynamic simulations covering 10 structural configurations and an unseen 4-story frame, an unexpected phenomenon emerges:

- The continuous directional cosine alignment transfers zero-shot ($\cos \theta \approx +0.80$ to $+0.85$, peaking at $+0.996$). The model reliably identifies which side of the structure has more damage.

- However, predicted damage separation collapses to $\sim 10^{-5}$ (compared to ground truth 0.4243), returning discrete attribution to 50.0% chance.
- A 72-model ablation scaling training from 12 to 100 epochs showed that training longer refines directional alignment but cannot break the magnitude collapse. Parameter dispersion across buildings creates an implicit regularization that compresses output magnitudes toward the mean.

3. Quantitative Benchmarks (Held-Out Test Set, N = 30)

Modality Suite	Channels	Fisher Sensitivity $\sqrt{I_{AB}}$	Attribution Accuracy	Exact Binomial p-value	95% Clopper-Pearson CI	Predicted Separation
S0 (Horizontal Accel)	3	5.62	50.0% (15/30)	$p = 1.000$	[31.3%, 68.7%]	0.000009
S2 (Horiz + Column Strain)	9	1344.18	50.0% (15/30)	$p = 1.000$	[31.3%, 68.7%]	0.000328
S1 (Horiz + Vert Accel)	9	1829.63	80.0% (24/30)	$p = 0.0014$	[61.4%, 92.3%]	0.1576
S4 (Multimodal Union)	15	2270.54	90.0% (27/30)	$p = 9.0 \times 10^{-6}$	[73.5%, 97.9%]	0.1671 (39.4% true)

4. Training-Budget Ablation Trajectory (Phase 7, P2 S4)

Epoch Budget	In-Distribution Acc	In-Distribution Cosine	Topological OOD Cosine	Damage Separation $\ \Delta \hat{d}\ _2$	Interpretation
12 Epochs	50.0% $\pm$ 0.0%	+0.467 $\pm$ 0.037	+0.489 $\pm$ 0.012	2.68×10^{-6}	Initial baseline; weak positive directional torque.
25 Epochs	50.0% $\pm$ 0.0%	+0.852 $\pm$ 0.115	+0.777 $\pm$ 0.158	7.68×10^{-6}	Steep rise in directional cosine alignment.
50 Epochs	50.0% $\pm$ 0.0%	+0.897 $\pm$ 0.038	+0.837 $\pm$ 0.090	1.23×10^{-5}	Peak mean directional alignment across seeds.
100 Epochs	50.0% $\pm$ 0.0%	+0.850 $\pm$ 0.205	+0.798 $\pm$ 0.199	1.83×10^{-5}	Orientation refined (peaks +0.996); separation remains collapsed.

5. Summary of Candidate Contributions & Research Interests

This project demonstrates my technical capability and research mindset:

- Structural Dynamics Modeling:** End-to-end parametric OpenSeesPy finite-element pipeline enforcing strict SI units and realistic seismic ground motions.
- Mathematical Rigor:** Derivation and implementation of noise-whitened sensitivity Jacobians and directional Fisher information to establish observability bounds.
- Novel ML Architecture:** Development of `DualStreamGFNO` combining temporal Fourier neural operators with structural graph convolutions, symmetry decomposition, and hierarchical damage heads.
- Scientific Honesty:** Transparent documentation of failure modes (the $\$S2\$$ learnability gap and cross-structure magnitude collapse) rather than cherry-picking isolated successes.

In your laboratory, I hope to extend this work to 3D structural frames with torsional coupling, group-equivariant neural operators ($\$C_2 / D_4\$$ GNNs), and multiscale gradient optimization.